

CS 4403 – Term Project Proposal

Can AI Simulate Reality?

A Data Mining Study of Generated vs Authentic Datasets

Team Members: Medha Madhub (medha.madhub@unb.ca), Fatima Asif (fatima.asif@unb.ca), Chanchal Preet Kaur (chanchalpreetk.1204@unb.ca)

GitHub Repository: <https://github.com/q62ur/CS4403-Term-Project>

Roles and Responsibilities

- **Data Collection:** Data sourcing, format verification, early cleaning
All members
- **Data Preprocessing:** Feature engineering, handling missing values, normalization
Lead: Fatima Asif; Contributors: Chanchal Preet Kaur, Medha Madhub
- **Modeling:** Clustering, classification, experimentation
Lead: Medha Madhub; Contributors: Fatima Asif, Chanchal Preet Kaur
- **Results and Analysis:** Interpretation, evaluation, visualization
Lead: Chanchal Preet Kaur; Contributors: Medha Madhub, Fatima Asif
- **Reporting:** Proposal, interim and final reports, editing and visuals
All members
- **GitHub Management:** Version control, meaningful commits, synchronization
All members

Communication Platforms

The group will communicate primarily through a **WhatsApp group** for quick coordination. Formal meetings will be conducted using **Microsoft Teams**, and the group will also meet **in person on campus** for discussions and progress updates.

Handling Disagreements

- **Open communication:** All members may explain their reasoning before decisions are made.
- **Data-driven decisions:** Evidence from datasets will guide choices whenever possible.
- **Compromise:** Discussions will continue until a mutually acceptable solution is reached.
- **Clear roles:** Responsibilities are assigned early to avoid confusion.
- **Regular meetings:** Issues will be addressed early to prevent escalation.
- **Decision rules:** Majority voting or comparative testing will be used if required.

Dataset Description

What dataset will be used and where will it be obtained?

We will use publicly available structured datasets from **Kaggle**, such as student performance or business analytics datasets. In addition, we will use **large language models (e.g., ChatGPT)** to generate synthetic datasets with similar variables and structure. All AI-generated data will be clearly labeled and evaluated separately to maintain academic integrity.

All datasets will be collected and generated by February 23rd.

Dataset Size Justification

The Kaggle datasets will contain **hundreds to several thousand records**, sufficient for statistical analysis, regression modeling, and comparative evaluation. The synthetic datasets will match the size and structure of the real datasets to ensure fair comparison.

Information and Insight to be Mined

We will analyze how closely AI-generated datasets resemble real-world data by comparing:

- Statistical distributions (mean, variance, correlations)
- Patterns and trends
- Model performance when trained on real vs synthetic data

The goal is to measure how accurately AI can replicate real data characteristics.

Proposed Use of Results

If synthetic datasets closely match real data, organizations with limited data access can use AI-generated data for testing, simulation, and model training. This can reduce cost, protect privacy, and improve efficiency while supporting decision-making.

Why This Project is Interesting

Many organizations face challenges related to data scarcity and privacy constraints. This project evaluates whether synthetic data can serve as a reliable alternative to real datasets. It is particularly relevant as AI-generated data becomes more common in analytics and predictive modeling.

What We Hope to Learn

We aim to determine:

- Whether AI-generated data can realistically replicate real-world datasets
- The strengths and weaknesses of synthetic data
- Its reliability for predictive modeling
- Situations where synthetic data may be a practical substitute for real data

Preliminary Timeline

- **Feb 12:** Submit Group Proposal
- **By Feb 23:** Select Kaggle datasets, generate synthetic data, confirm structure
- **Feb 24 – Mar 9:** Data cleaning, analysis, modeling, interim report preparation
- **Mar 10:** Submit Interim Report
- **Mar 11 – Mar 31:** Model refinement and comparative analysis
- **Apr 1 – Apr 6:** Final report and presentation preparation
- **Apr 7:** Project Presentation